

CS-412 Operating Systems

Assignment

Date: April 16, 2026

Marks: 20 (5*4)

Please Note.

Originality will be the most important criteria for submission of assignment. The assignment will be checked by the standard plagiarism tool of JNU and if found more than 15 % similarity, it will be rejected. Evaluation will be based on proper understanding of concepts.

Answers should be submitted in a single file in pdf format (either hand-written or computer typed). The assignment is supposed to be submitted by April 28, 2026 (Hard deadline) at (ratnesh@jnu.ac.in) with Name and course name in Subject line)

THERE IS NO PAGE LIMIT CONSTRAINT.

1. The experimental Synthesis operating system has an assembler incorporated in the kernel. To optimize system-call performance, the kernel assembles routines within kernel space to minimize the path that the system call must take through the kernel. This approach is the antithesis of the layered approach, in which the path through Kernel is extended to make building the operating system easier. Discuss the pros and cons of the synthesis approach to kernel design and system performance optimization.
2. Linux does not distinguish between processes and threads. Instead, Linux treats both in same way, allowing a task to be more akin to a process or a thread depending on the set of flags passed to the system call. However, other operating systems, such as Windows, treat processes and threads differently. Typically, such systems use a notation in which the data structure for a process contains pointers to the separate threads belonging to the process. Contrast these two approaches with suitable example demonstrations for modelling processes and threads within the kernel.
3. Five philosophers sit around a table, but the resources and the philosophers themselves are no longer identical. Instead of simple chopsticks, there are five tools with different "Weights" (W_1 to W_5). Each philosopher P_i has a "Strength" attribute (S_i). To eat, P_i must hold two adjacent utensils whose combined weight $W_{left} + W_{right} \leq S_i$. If the tools are too heavy, the philosopher cannot pick them up. Explain all the constraints and implement the solution using any programming language in a strictly non-blocking, lock-free manner and using only atomic primitives. Solution must prevent "Strong" philosopher from starving "Weak" neighbours.
4. Design a Segmentation + Paging hybrid mapping where Logical Address= (Segment#, Offset) $\rightarrow$ (Page#, Frame). Demonstrate the conditions under which hybrid mapping reduces fragmentation (either external or internal) and where it increases translation overhead. Discuss the space complexity of page tables and trade-off between Table size and look-up time.